An Extension of the Generalized Riemann Functional Equation

Jose Risomar Sousa
ORCID: 0009-0002-7779-0808

June 18, 2025

Abstract

An extension of the generalized Riemann functional equation from the literature that holds for $b > 0$ (unlike the one in the literature, which holds for $0 \leq \Re(b) \leq 1$) is introduced. The insight that led to this new relation came from a new formula for the zeta function created herein that implies the Riemann functional equation – when the same process is replicated with a new formula for the Hurwitz zeta function, the extended relation is obtained. A few minor developments that stem from new formulae introduced previously are also discussed.

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1 Introduction

This paper presents new developments that stem from new formulae introduced previously for generalized harmonic progressions[4] and their limits[5].

Among these developments is a generalization of a transformation of the $H_k(n)$ formula that had not been discovered before[1], and one more integral representation for the Riemann zeta function, $\zeta(k)$, that only holds at the positive integers greater than one. A demonstration of how this new zeta function formula can be extended from the positive integers to the right half-plane, $\Re(k) > 1$, is provided.

More strikingly, a closer inspection of this extended Riemann zeta function formula led to an unexpected discovery – it was found to be just a reformulation of the Riemann functional equation. This in turn provided insight into the possibility of replicating the method with a new formula for the Hurwitz zeta function[2], $\zeta(k, b)$, which was first introduced in [5] and then extended in [7]. As a result, an extension of the generalized Riemann functional equation from the literature was found, though it only applies to positive b – unlike the relation from the literature, which allows for complex b with $0 \leq \Re(b) \leq 1$.

According to reference [1], the relation below,

$$\zeta(1-k,b) = \frac{\Gamma(k)}{(2\pi)^k}\left(e^{-k\,\boldsymbol{i}\,\pi/2}\sum_{n=1}^{\infty}\frac{e^{2\pi\boldsymbol{i}\,b\,n}}{n^k} + e^{k\,\boldsymbol{i}\,\pi/2}\sum_{n=1}^{\infty}\frac{e^{-2\pi\boldsymbol{i}\,b\,n}}{n^k}\right),$$

holds if $\Re(k) > 1$ and $0 \leq \Re(b) \leq 1$.

If this relation is transformed under $k \mapsto -k+1$, and the infinite series are replaced by their analytic continuation, the polylog function, the new relation,

$$\zeta(k,b) = -\boldsymbol{i}\,(2\pi)^{k-1}\,\Gamma(1-k)\left(e^{k\,\boldsymbol{i}\,\pi/2}\,\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right) - e^{-k\,\boldsymbol{i}\,\pi/2}\,\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)\right), \tag{1}$$

holds for all complex k (except for singularities) and thus generalizes the Riemann functional equation (which is the case $b = 1$).

The new functional equation created herein is nearly identical to the literature, except for the addition of a new term that is a function of k and b, and holds when b is a positive real number (except for occasional singularities – though it holds at the limit when improper).

2 Generalized harmonic progressions

In reference [3], new formulae were created for the generalized harmonic numbers, $H_k(n)$, and varied depending on whether k was odd or even. In reference [4], the same approach was

[1] See section (2.1) for details.

used to produce formulae for a more general sum referred to as generalized harmonic progression, $\mathrm{HP}_k(n)$.

For any integers a, b and $k \geq 1$,

$$\sum_{j=1}^{n} \frac{1}{(aj+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(an+b)^k} + \frac{\boldsymbol{i}\,(2\pi\boldsymbol{i})^k}{2} \int_0^1 \sum_{j=0}^{k} \frac{B_j\,(1-u)^{k-j}}{j!(k-j)!} \left(e^{2\pi\boldsymbol{i}(an+b)u} - e^{2\pi\boldsymbol{i}bu}\right) \cot \pi a u \, du, \quad (2)$$

where the right-hand side can be well-defined even when there are singularities on the left.

2.1 Generalized harmonic numbers

For the generalized harmonic numbers, as seen in [4], a single formula that does not depend on parity is,

$$\sum_{j=1}^{n} \frac{1}{j^k} = \frac{1}{2n^k} + \frac{\boldsymbol{i}\,(2\pi\boldsymbol{i})^k}{2} \int_0^1 \sum_{j=0}^{k} \frac{B_j\,(1-u)^{k-j}}{j!(k-j)!} \left(e^{2\pi\boldsymbol{i}nu} - 1\right) \cot \pi u \, du, \quad (3)$$

which is simply a particular case of formula (2).

Paper [3] mentioned how for the odd cases the formulae could be transformed based on a new integral representation for the zeta function at the odd integers (which stems from the formula itself). For example, for the case $k = 3$ one had the following,

$$\sum_{j=1}^{n} \frac{1}{j^3} = \frac{1}{2n^3} + \zeta(3) - \frac{\pi^3}{12} \int_0^1 \left(u - u^3\right) \cos \pi n(1-u) \tan \frac{\pi u}{2} \, du,$$

although a generalization of this result was not produced.

But with formula (3) this is now straightforward. A simple investigation reveals that in this case the generalization is given by,

$$\sum_{j=1}^{n} \frac{1}{j^k} = \frac{1}{2n^k} + \zeta(k) + \frac{\boldsymbol{i}\,(2\pi\boldsymbol{i})^k}{2} \int_0^1 \sum_{j=0}^{k-1} \frac{B_j\,(1-u)^{k-j}}{j!(k-j)!} e^{2\pi\boldsymbol{i}nu} \cot \pi u \, du, \quad (4)$$

that is, the harmonic numbers of order k are a function of $\zeta(k)$ (and vice-versa).

2.2 Zeta at the positive integers

From the limits of the formulae created in [3] new integral representations for the Riemann zeta function at the positive odd integers (greater than one) were derived, such as,

$$\zeta(2k+1) = \frac{(-1)^k \pi^{2k+1}}{2} \int_0^1 \sum_{j=0}^{k} \frac{B_{2j}\,(2-2^{2j})\,u^{2k+1-2j}}{(2j)!(2k+1-2j)!} \tan \frac{\pi u}{2} \, du, \quad (5)$$

and,

$$\zeta(2k+1) = -\frac{(-1)^k (2\pi)^{2k+1}}{2} \int_0^1 \sum_{j=0}^{k} \frac{B_{2j}\,(2-2^{2j})\,u^{2k+1-2j}}{(2j)!(2k+1-2j)!} \cot \pi u \, du \tag{6}$$

The limits of the real and imaginary parts of each individual component of the integral in (3) were initially deduced through experimentation[2] and their validity was confirmed later by the integral representation derived for zeta at the odd integers from (6) (which stems from a formula for $H_k(n)$ created using an initial equation other than the exponential function, per reference [3]). This was first done in [5] and the statements were later refined in [6].

If k is real, the limit of the imaginary part can be stated as,

$$\lim_{n\to\infty} \int_0^1 (1-u)^k \sin 2\pi n u \cot \pi u \, du = \begin{cases} 1, & \text{if } k = 0 \\ 1/2, & \text{if } k > 0, \end{cases} \tag{7}$$

and the limit of the real part as,

$$\int_0^1 (1-u)^k (\cos 2\pi n u - 1) \cot \pi u \, du \sim -\frac{H(n)}{\pi} + \int_0^1 (u^k - u) \cot \pi u \, du, \tag{8}$$

which only holds for positive real k. If $k = 0$ and n is a positive integer, the integral on the left-hand side is zero, while the integral to the right diverges.

Paper [4] did not show how the values of the zeta function at the positive integers (greater than one) can be derived from (3), but from (4) this is now possible. First, note that for positive integer n,

$$\int_0^1 \left(e^{2\pi \boldsymbol{i} n u} - 1\right) \cot \pi u \, du = \boldsymbol{i}, \tag{9}$$

and therefore (4) can be rewritten as:

$$\sum_{j=1}^{n} \frac{1}{j^k} = \frac{1}{2n^k} - \frac{B_k\,(2\pi \boldsymbol{i})^k}{2\,k!} + \frac{\boldsymbol{i}\,(2\pi \boldsymbol{i})^k}{2} \int_0^1 \left(\sum_{j=0}^{k} \frac{B_j\,(1-u)^{k-j}}{j!(k-j)!} - \frac{B_k}{k!} \right) \left(e^{2\pi \boldsymbol{i} n u} - 1\right) \cot \pi u \, du,$$

Through the aforementioned limits, (7) and (8), if k is a positive integer greater than one then,

$$\begin{aligned} \zeta(k) &= -\frac{B_k\,(2\pi \boldsymbol{i})^k}{2\,k!} + \frac{\boldsymbol{i}\,(2\pi \boldsymbol{i})^k}{2} \int_0^1 \sum_{j=0}^{k-1} \frac{B_j\,u^{k-j}}{j!(k-j)!} \cot \pi u \, du \\ &= \frac{\boldsymbol{i}\,(2\pi \boldsymbol{i})^k}{2} \int_0^1 \left(\frac{B_k}{k!} e^{2\pi \boldsymbol{i} n u} - \sum_{j=0}^{k} \frac{B_j\,(1-u)^{k-j}}{j!(k-j)!} \right) \cot \pi u \, du, \end{aligned} \tag{10}$$

where the rightmost formula was obtained from equations (3) and (4) and, again, n is any positive integer. Oddly, these formulae hold even for $k = 0$.

[2] Though they are now proved.

2.3 Extending $H_k(n)$

How can one obtain the extension of these formulae from positive integer to complex k?

The summation within the integral of formula (2) can be extended to complex k as follows,

$$\sum_{j=0}^{k+1} \frac{B_j\, u^{k+1-j}}{j!(k+1-j)!} = \sum_{q=-1}^{k} \frac{B_{q+1} u^{k-q}}{(q+1)!(k-q)!} = \frac{u^{k+1}}{(k+1)!} - \frac{u^k}{k!} - \sum_{q=0}^{k} \frac{\zeta(-q)\, u^{k-q}}{q!(k-q)!},$$

since for integer q:

$$\frac{B_{q+1}}{q+1} = \begin{cases} -1/2, & \text{if } q = 0 \\ -\zeta(-q), & \text{if } q > 0 \end{cases},$$

Therefore, using an identity for $\zeta(-k, u+1)$ from [7] one has,

$$\sum_{j=0}^{k+1} \frac{B_j\, u^{k+1-j}}{j!(k+1-j)!} = -\frac{u^k}{k!} - \frac{1}{k!}\left(-\frac{u^{k+1}}{k+1} + \sum_{q=0}^{k} \binom{k}{q} \zeta(-q)\, u^{k-q}\right)$$
$$= -\frac{1}{k!}\left(u^k + \zeta(-k, u+1)\right) = -\frac{\zeta(-k,u)}{k!},$$

and if the rightmost function is differentiated once with respect to u then:

$$\sum_{j=0}^{k} \frac{B_j\, u^{k-j}}{j!(k-j)!} = -\frac{\zeta(-k+1,u)}{(k-1)!} \tag{11}$$

This extended summation in turn implies,

$$\sum_{j=1}^{n} \frac{1}{j^k} = \frac{1}{2n^k} - \frac{\boldsymbol{i}\,(2\pi\boldsymbol{i})^k}{2\,\Gamma(k)} \int_0^1 \zeta(-k+1, 1-u)\left(e^{2\pi\boldsymbol{i}\,nu} - 1\right) \cot \pi u\, du$$
$$= \frac{1}{2n^k} + \zeta(k) - \frac{\boldsymbol{i}\,(2\pi\boldsymbol{i})^k}{2\,\Gamma(k)} \int_0^1 \left(\zeta(-k+1, 1-u) - \zeta(-k+1)\right) e^{2\pi\boldsymbol{i}\,nu} \cot \pi u\, du, \tag{12}$$

where the first formula holds for $\Re(k) > 0$, when equation (3) is used, and the second formula holds for $\Re(k) > 1$, when equation (4) is used.

3 The extended zeta formula

Through the extended summation from (11), the Riemann zeta function formula in (10) can be extended from the non-negative integers to $\Re(k) > 1$:

$$\zeta(k) = \frac{(2\pi\boldsymbol{i})^k\, \zeta(-k+1)}{2\,\Gamma(k)} - \frac{\boldsymbol{i}\,(2\pi\boldsymbol{i})^k}{2\,\Gamma(k)} \int_0^1 \left(\zeta(-k+1, u) - \zeta(-k+1)\right) \cot \pi u\, du \tag{13}$$

From this equation, the Riemann functional equation can be deduced as follows. Since for real k the imaginary part on the right-hand side must be zero, it can be used to deduce a closed form for the real-valued integral,

$$\int_0^1 (\zeta(-k+1,u) - \zeta(-k+1)) \cot \pi u \, du = \tan \frac{k\,\pi}{2}\, \zeta(-k+1),$$

and the above formula, when replaced into the real part of the initial equation, leads to a restated Riemann functional equation:

$$\zeta(k) = \frac{(2\pi)^k}{2\,\Gamma(k)}\, \zeta(-k+1) \left(\cos \frac{k\,\pi}{2} - \cos \frac{(k+1)\pi}{2} \tan \frac{k\,\pi}{2} \right) = \frac{(2\pi)^k}{2\,\Gamma(k)}\, \zeta(-k+1) \sec \frac{k\,\pi}{2}$$

This must explain why the analytic continuation of the zeta function appears in the new formula.

3.1 The extended Hurwitz zeta formula

The prior result was the insight that led to the discovery of an extension of the generalized Riemann functional equation from the literature, whose derivation requires both parameters, k and b, to be real.

Throughout this section, let b be a positive non-integer number, unless otherwise noted. The extended Hurwitz zeta function formula from reference [7], which holds in the right half-plane $\Re(k) > 1$, is:

$$\begin{aligned} \zeta(k,b) = \sum_{j=0}^{\infty} \frac{1}{(j+b)^k} &= \frac{1}{2\,b^k} + \frac{(2\pi \boldsymbol{i})^k}{2\,\Gamma(k)}\, \mathrm{Li}_{-k+1}\left(e^{-2\pi \boldsymbol{i}\, b}\right) \\ &\quad - \frac{\boldsymbol{i}\,(2\pi \boldsymbol{i})^k}{2\,\Gamma(k)} \int_0^1 \left(e^{-2\pi \boldsymbol{i}\, b\, u}\, \Phi(e^{-2\pi \boldsymbol{i}\, b}, -k+1, u) - \mathrm{Li}_{-k+1}\left(e^{-2\pi \boldsymbol{i}\, b}\right) \right) \cot \pi u \, du \end{aligned} \quad (14)$$

If the shift is $-b$ though, the picture gets more complicated and the above formula does not hold (the real part is right, but not the imaginary part).

To understand why, it is necessary to analyze what happens when the shift is negative. By definition, a real power of a positive number is always a positive number[3], but if the base is negative, the result can be a non-real number. For example, if $-1 < -b < 0$, then all the terms of the Hurwitz zeta series are positive and hence well-behaved, except term $j = 0$. For any other b, only the terms of the series that have $j - b < 0$ contribute to the imaginary part of $\zeta(k, -b)$, that is:

$$\Im(\zeta(k,-b)) = \Im\left(\sum_{j=0}^{\lfloor b \rfloor} \frac{1}{(j-b)^k} \right) \quad (15)$$

[3]When the exponentiation is multi-valued, for example, $4^{1/2} = \pm 2$, the positive root is taken by definition.

Fortunately, it is not hard to fix the formula (14) when the shift is $-b$, as a simple observation of the patterns reveals that, surprisingly, the difference can be attributed to the imaginary part of the terms that become negative, as in the below:

$$\zeta(k,-b)=\sum_{j=0}^{\infty}\frac{1}{(j-b)^k}=\frac{1}{2(-b)^k}+\boldsymbol{i}\,\Im\left(\frac{1}{(-b)^k}+2\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)+\frac{(2\pi\boldsymbol{i})^k}{2\,\Gamma(k)}\,\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)$$
$$-\frac{\boldsymbol{i}\,(2\pi\boldsymbol{i})^k}{2\,\Gamma(k)}\int_0^1\left(e^{2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{2\pi\boldsymbol{i}\,b},-k+1,u)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\cot\pi u\,du \quad (16)$$

3.2 Reflection symmetry relations

When the two equations, (14) and (16), are added together or subtracted from each other, each part becomes either purely real or purely imaginary regardless of the value of b. This is essential for the method to work since the idea is to form a system of equations, so that the unknown part, the integral, can be replaced by an expression involving well-known functions, leading to a closed form.

The difference between the two reflection formulae is below,

$$\zeta(k,b)-\zeta(k,-b)=\frac{1}{2\,b^k}-\frac{1}{2(-b)^k}-\boldsymbol{i}\,\Im\left(\frac{1}{(-b)^k}+2\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)$$
$$+\frac{(2\pi\boldsymbol{i})^k}{2\,\Gamma(k)}\left(\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)$$
$$-\frac{\boldsymbol{i}\,(2\pi\boldsymbol{i})^k}{2\,\Gamma(k)}\int_0^1\left(e^{-2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{-2\pi\boldsymbol{i}\,b},-k+1,u)-e^{2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{2\pi\boldsymbol{i}\,b},-k+1,u)\right.$$
$$\left.-\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\cot\pi u\,du, \quad (17)$$

and the corresponding formula for the addition is not shown since it is trivially similar. These formulae allow the real and imaginary parts to be separated.

4 Difference formulae

This section was created to organize the calculations better and make them easier to follow. A system of equations is formed with the purpose of replacing the integral with an equivalent closed form.

4.1 The real part

The real part of the difference, equation (17), is:

$$\begin{aligned}\zeta(k,b)-\Re(\zeta(k,-b)) &= \frac{1}{2\,b^k}-\Re\left(\frac{1}{2(-b)^k}\right)\\ &\quad+\frac{(2\pi)^k}{2\,\Gamma(k)}\boldsymbol{i}\sin\frac{k\,\pi}{2}\left(\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\\ &\quad-\frac{(2\pi)^k}{2\,\Gamma(k)}\boldsymbol{i}\sin\frac{(k+1)\pi}{2}\int_0^1\Big(e^{-2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{-2\pi\boldsymbol{i}\,b},-k+1,u)-e^{2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{2\pi\boldsymbol{i}\,b},-k+1,u)\\ &\qquad-\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\Big)\cot\pi u\,du\end{aligned}\tag{18}$$

4.2 The imaginary part

The imaginary part of the difference is:

$$\begin{aligned}-\boldsymbol{i}\,\Im(\zeta(k,-b)) &= -\boldsymbol{i}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=0}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)\\ &\quad+\frac{(2\pi)^k}{2\,\Gamma(k)}\cos\frac{k\,\pi}{2}\left(\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\\ &\quad-\frac{(2\pi)^k}{2\,\Gamma(k)}\cos\frac{(k+1)\pi}{2}\int_0^1\Big(e^{-2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{-2\pi\boldsymbol{i}\,b},-k+1,u)-e^{2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{2\pi\boldsymbol{i}\,b},-k+1,u)\\ &\qquad-\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\Big)\cot\pi u\,du\end{aligned}$$

Notice the cancellation of $\Im\zeta(k,-b)$ with its respective equivalent formula from (15). The equation can therefore be simplified as below:

$$\begin{aligned}&\frac{(2\pi)^k}{2\,\Gamma(k)}\cos\frac{(k+1)\pi}{2}\int_0^1\Big(e^{-2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{-2\pi\boldsymbol{i}\,b},-k+1,u)-e^{2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{2\pi\boldsymbol{i}\,b},-k+1,u)\\ &\qquad-\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\Big)\cot\pi u\,du=\\ &-\boldsymbol{i}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)+\frac{(2\pi)^k}{2\,\Gamma(k)}\cos\frac{k\,\pi}{2}\left(\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\end{aligned}\tag{19}$$

4.3 The closed form

When the closed form of the integral from (19) is replaced into equation (18), the below is obtained:

$$\begin{aligned}\zeta(k,b)-\Re(\zeta(k,-b)) &= \frac{1}{2\,b^k}-\Re\left(\frac{1}{2(-b)^k}\right)\\ &\quad+\frac{(2\pi)^k}{2\,\Gamma(k)}\boldsymbol{i}\sin\frac{k\,\pi}{2}\left(\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\\ -\boldsymbol{i}\tan\frac{(k+1)\pi}{2}&\left(-\boldsymbol{i}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)+\frac{(2\pi)^k}{2\,\Gamma(k)}\cos\frac{k\,\pi}{2}\left(\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\right)\end{aligned}$$

It can be simplified using the below identity,

$$\boldsymbol{i}\sin\frac{k\,\pi}{2}-\boldsymbol{i}\tan\frac{(k+1)\pi}{2}\cos\frac{k\,\pi}{2}=\boldsymbol{i}\csc\frac{k\,\pi}{2}, \tag{20}$$

which then gives the final relation for the difference:

$$\begin{aligned}\zeta(k,b)-\Re(\zeta(k,-b)) &= \frac{1}{2\,b^k}-\Re\left(\frac{1}{2(-b)^k}\right)\\ &\quad+\frac{(2\pi)^k}{2\,\Gamma(k)}\boldsymbol{i}\csc\frac{k\,\pi}{2}\left(\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\\ &\quad-\tan\frac{(k+1)\pi}{2}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)\end{aligned} \tag{21}$$

5 Addition formulae

Like before, this section aims to organize better the calculations of the system of equations obtained by adding the reflection formulae together.

5.1 The real part

The real part of the addition is:

$$\begin{aligned}\zeta(k,b)+\Re(\zeta(k,-b)) &= \frac{1}{2\,b^k}+\Re\left(\frac{1}{2(-b)^k}\right)\\ &\quad+\frac{(2\pi)^k}{2\,\Gamma(k)}\cos\frac{k\,\pi}{2}\left(\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\\ -\frac{(2\pi)^k}{2\,\Gamma(k)}\cos\frac{(k+1)\pi}{2}&\int_0^1\left(e^{-2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{-2\pi\boldsymbol{i}\,b},-k+1,u)+e^{2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{2\pi\boldsymbol{i}\,b},-k+1,u)\right.\\ &\left.\quad-\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\cot\pi u\,du\end{aligned} \tag{22}$$

5.2 The imaginary part

Like in the corresponding section of the difference formulae, this equation can be simplified as:

$$\frac{(2\pi)^k}{2\,\Gamma(k)}\boldsymbol{i}\sin\frac{(k+1)\pi}{2}\int_0^1\Big(e^{-2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{-2\pi\boldsymbol{i}\,b},-k+1,u)+e^{2\pi\boldsymbol{i}\,b\,u}\,\Phi(e^{2\pi\boldsymbol{i}\,b},-k+1,u)$$
$$-\operatorname{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-\operatorname{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\Big)\cot\pi u\,du=$$
$$\boldsymbol{i}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)+\frac{(2\pi)^k}{2\,\Gamma(k)}\boldsymbol{i}\sin\frac{k\,\pi}{2}\left(\operatorname{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\operatorname{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\tag{23}$$

5.3 The closed form

When the closed form of the integral from (23) is replaced into equation (22), the below is obtained:

$$\zeta(k,b)+\Re(\zeta(k,-b))=\frac{1}{2\,b^k}+\Re\left(\frac{1}{2(-b)^k}\right)$$
$$+\frac{(2\pi)^k}{2\,\Gamma(k)}\cos\frac{k\,\pi}{2}\left(\operatorname{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\operatorname{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)$$
$$\boldsymbol{i}\cot\frac{(k+1)\pi}{2}\left(\boldsymbol{i}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)+\frac{(2\pi)^k}{2\,\Gamma(k)}\boldsymbol{i}\sin\frac{k\,\pi}{2}\left(\operatorname{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\operatorname{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)\right)$$

This one can also be simplified through the below identity,

$$\cos\frac{k\,\pi}{2}+\boldsymbol{i}\cot\frac{(k+1)\pi}{2}\,\boldsymbol{i}\sin\frac{k\,\pi}{2}=\sec\frac{k\,\pi}{2},\tag{24}$$

which then yields this final relation for the addition:

$$\zeta(k,b)+\Re(\zeta(k,-b))=\frac{1}{2\,b^k}+\Re\left(\frac{1}{2(-b)^k}\right)$$
$$+\frac{(2\pi)^k}{2\,\Gamma(k)}\sec\frac{k\,\pi}{2}\left(\operatorname{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)+\operatorname{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right)$$
$$-\cot\frac{(k+1)\pi}{2}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)\tag{25}$$

6 Extended functional equation

To obtain a simpler relation, formulae (21) and (25) are combined, with a few of the parts canceling.

The final relation is further simplified by means of the following identities,

$$\sec\frac{k\,\pi}{2}+\boldsymbol{i}\csc\frac{k\,\pi}{2}=2\,\boldsymbol{i}\,\frac{e^{-k\,\boldsymbol{i}\,\pi/2}}{\sin k\,\pi},$$
$$\sec\frac{k\,\pi}{2}-\boldsymbol{i}\csc\frac{k\,\pi}{2}=-2\,\boldsymbol{i}\,\frac{e^{k\,\boldsymbol{i}\,\pi/2}}{\sin k\,\pi},$$
$$\tan\frac{(k+1)\pi}{2}+\cot\frac{(k+1)\pi}{2}=-2\csc k\,\pi$$

which gives the relation:

$$\zeta(k,b)=\frac{1}{2\,b^k}+\frac{1}{\sin k\,\pi}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)$$
$$+\frac{\boldsymbol{i}\,(2\pi)^k}{2\,\Gamma(k)\,\sin k\,\pi}\left(e^{-k\,\boldsymbol{i}\,\pi/2}\,\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)-e^{k\,\boldsymbol{i}\,\pi/2}\,\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)\right),$$

Finally, Euler's reflection formula,

$$\frac{1}{2\,\Gamma(k)\,\sin k\,\pi}=\frac{\Gamma(1-k)}{2\pi},$$

leads to a relation that resembles the literature more closely[4],

$$\zeta(k,b)=\frac{1}{2\,b^k}+\frac{1}{\sin k\,\pi}\,\Im\left(\frac{1}{2(-b)^k}+\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(j-b)^k}\right)$$
$$-\boldsymbol{i}\,(2\pi)^{k-1}\Gamma(1-k)\left(e^{k\,\boldsymbol{i}\,\pi/2}\,\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)-e^{-k\,\boldsymbol{i}\,\pi/2}\,\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)\right),$$

which holds for all real k and all positive real b, except for the singularities at integer k (all of which can be resolved with limits, except $k=1$).

This new relation only extends the domain of the generalized functional equation from the literature when b is real (from $0\leq b\leq 1$ to $b\geq 0$). The proof requires (or implies) the below identity,

$$\Im\left(\frac{1}{(-b)^k}\right)=-\frac{\sin k\,\pi}{b^k},\ \text{if } b>0, \tag{26}$$

which in turn provides the insight for a neat simplification of the previous formula. Hence, if $k\neq 1$ is complex and b is a positive real number then:

$$\zeta(k,b)=-\sum_{j=1}^{\lfloor b\rfloor}\frac{1}{(b-j)^k}$$
$$-\boldsymbol{i}\,(2\pi)^{k-1}\Gamma(1-k)\left(e^{k\,\boldsymbol{i}\,\pi/2}\,\mathrm{Li}_{-k+1}\left(e^{2\pi\boldsymbol{i}\,b}\right)-e^{-k\,\boldsymbol{i}\,\pi/2}\,\mathrm{Li}_{-k+1}\left(e^{-2\pi\boldsymbol{i}\,b}\right)\right) \tag{27}$$

The Riemann functional equation is a particular case of this new functional equation, as it can be verified by making $b=1$ (the relation also holds for positive integer b, as long as the singularity that appears in the summation is removed).

[4] As seen in equation (1).

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